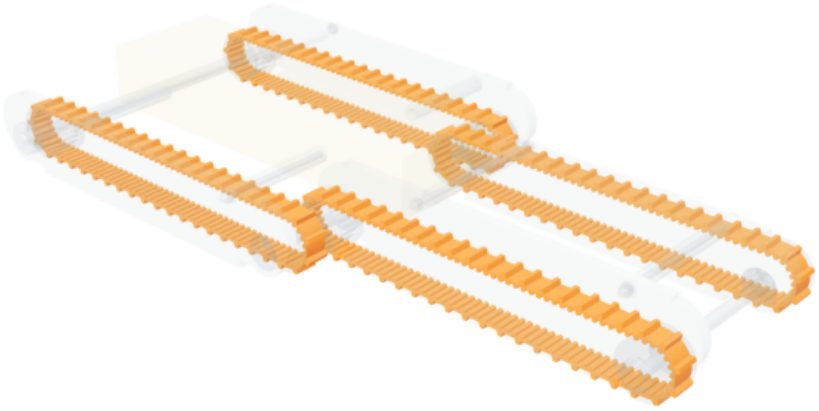
Track



Quantity: 4 (2 per segment)

MAKE: 3D print

What it does

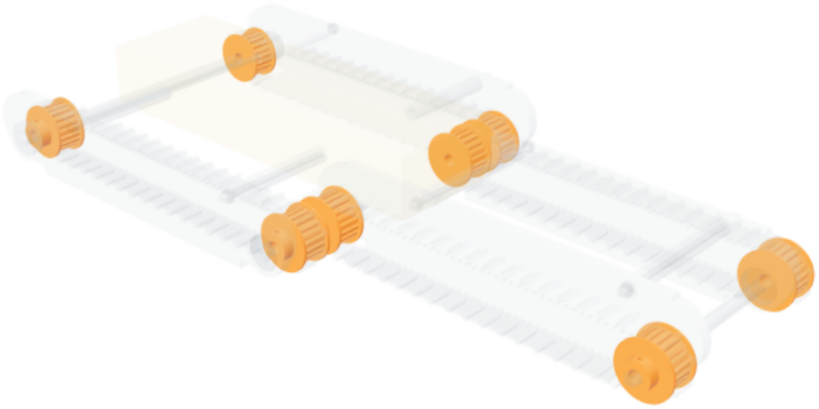
The belt itself, printed in flexible TPU. Its inside has 110 teeth that sit in the grooves of the toothed wheels, so it cannot slip. Its outside has 45 cleats spaced so that five of them span one ES1 step: every step edge lands between two cleats and the track grips the stair like a gear on a rack.

Where to get it

Fab Lab E4-02-06, Tiertime UP300 (it has the TPU extruder). File stl/track_TPU_x4.stl. Print standing as a ring, no supports, 0.2 mm layers, 100% infill, 20-30 mm/s. About 3 h and 32 g each. Filament: TPU 95A, Fab Lab stock or Element14 SG (sg.element14.com, Multicomp Pro TPU 1.75 mm, 1-2 days).

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-01 in the package PDF.

Toothed wheel (timing pulley)



Quantity: 8 (4 per segment)

BUY: MISUMI Singapore

What it does

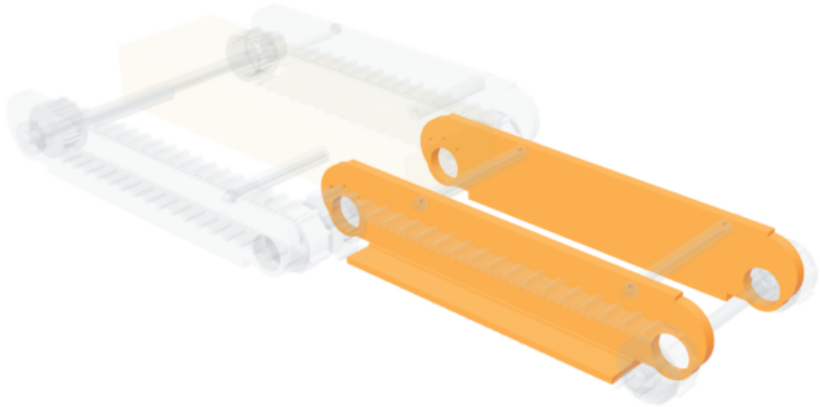
The wheels the track runs on, two per side per segment. HTD 5M profile, 20 teeth, with a thin flange each side so the track cannot walk off. Each is locked to its axle by two M4 set screws onto a filed flat, so when the middle axle turns, all four of its wheels turn and both tracks move.

Where to get it

sg.misumi-ec.com > Timing Pulleys HTD 5M (economy series). Choose 20 teeth, belt width 15, bore 8, flanged. Family code C-HTPA20H5M150. Order 16 x M4 set screws with them (not included). Ships from Singapore stock in about 1 day; allow 2-5 working days. About S$20 each.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-06 in the package PDF.

Side plate, front segment



Quantity: 2 (1 left + 1 right, mirror images)

MAKE: 3D print, or CNC at the Professional Work

What it does

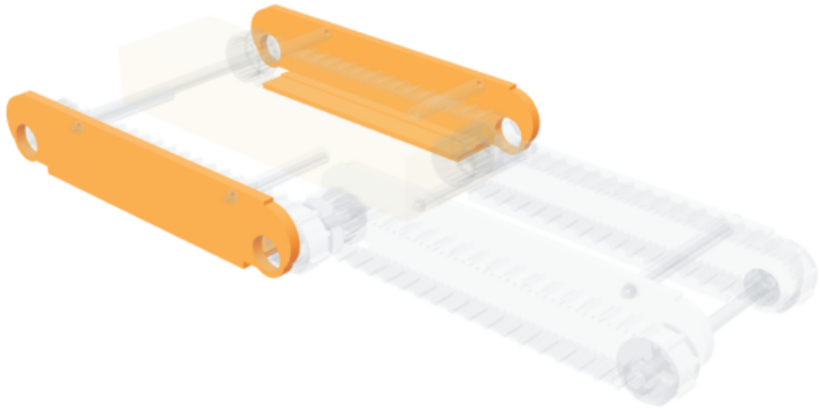
The frame of the front segment. Each plate holds the two bearings the axles turn in, carries the slide rail that supports the track's lower run, has the two cross-bar holes, and on this plate the three holes for the hinge sector gear. The front plates sit inboard (z 41-48) so the front segment can swing between the rear segment's wheels.

Where to get it

Fab Lab, PETG, flat on the bed, 0.2 mm layers, 4 walls, 30% infill, no supports, about 5 h each; needs a 235 mm bed (UP300 on the diagonal, or UP600). Files stl/side_plate_front_L.stl and _R.stl. Or: job request to the Professional Workshop (E3-01-06) to CNC-mill from 7 mm aluminium using sheet TS-02 and step/side_plate_front.step.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-02 in the package PDF.

Side plate, rear segment



Quantity: 2 (1 left + 1 right)

MAKE: 3D print, or CNC

What it does

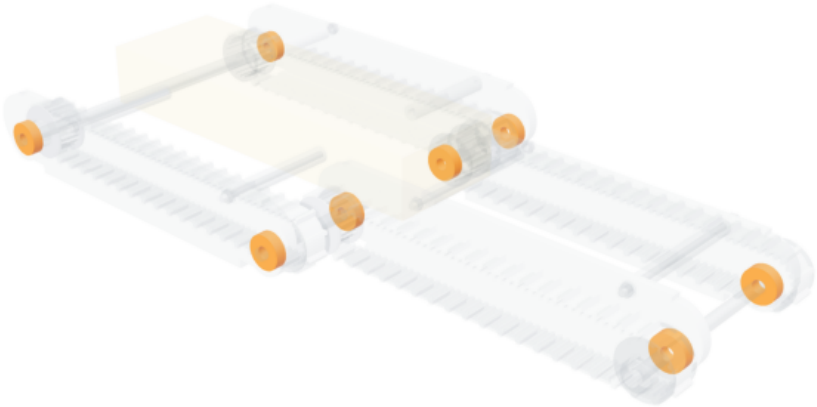
Same job as the front plate but for the rear segment: bearings, slide rail, cross-bar holes. These plates sit outboard (z 105-112), outside everything, and the box is clamped between them by the cross bars. No sector gear holes.

Where to get it

As the front plate. Files stl/side_plate_rear_L.stl and _R.stl, or sheet TS-03 and step/side_plate_rear.step for the workshop.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-03 in the package PDF.

Ball bearing 608ZZ



Quantity: 10 (8 in the plates + 2 for the hinge cross sh

BUY: RS Singapore

What it does

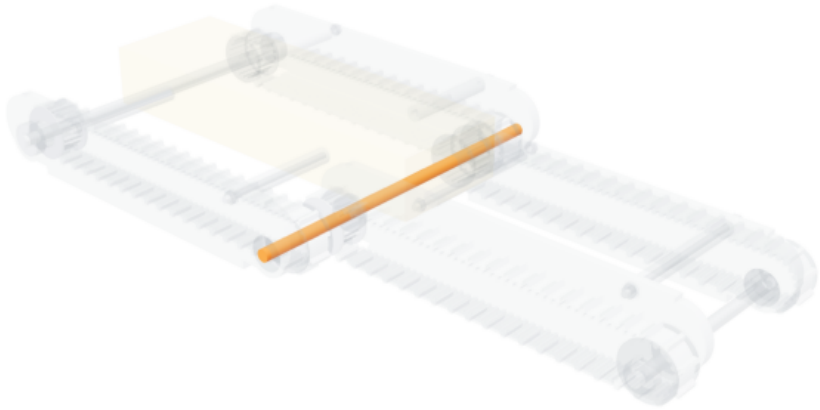
Pressed into the Ø22.2 pocket at each end of each plate. The axle runs through its 8 mm bore, so the axle spins freely while the plate stays still. 8 x 22 x 7 mm, shielded both sides (ZZ) against dust. The same bearing as in a skateboard wheel.

Where to get it

sg.rs-online.com, RS PRO 608-2Z/C3, pack of 5, stock no. 261-2603, S$14.82 per pack ex GST; buy 2 packs. 657 in stock, next working day if ordered before 3:30 pm. Alternatives on the approved list: Wheeler Mectrade, any SKF distributor.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-07 in the package PDF.

Middle axle (the hinge)



Quantity: 1

BUY rod, cut to 230 mm

What it does

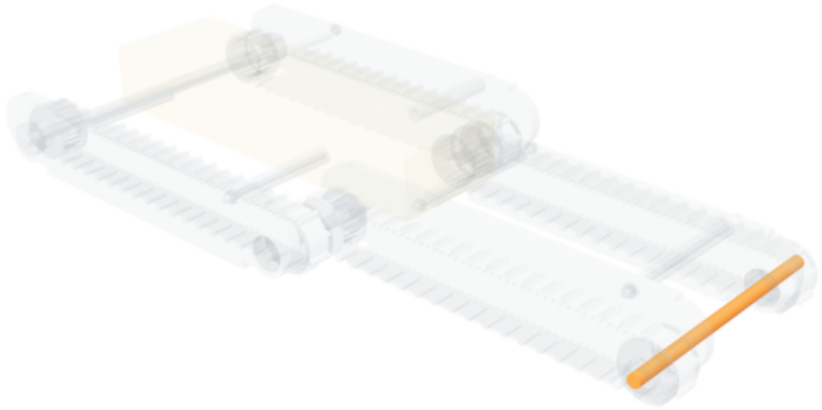
The shaft both segments share. It carries four toothed wheels (two front-segment, two rear-segment) and runs in all four plates' bearings. The drive motor turns it, so one motor drives every track. The front segment also pivots about it: it is the hinge. Flats are filed where the set screws land.

Where to get it

Ø8 steel rod. First choice: Design Lab spare rod (EA-03-04), free. Else sg.rs-online.com > linear shafts, 8 mm x 400 mm (about S$37, hardened: grind the flats with a Dremel), or soft mild-steel rod from a hardware shop on the vendor list (Teck Huat, Hillside), about S$5. Cut square, deburr.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-04 in the package PDF.

Front axle



Quantity: 1

BUY rod, cut to 160 mm

What it does

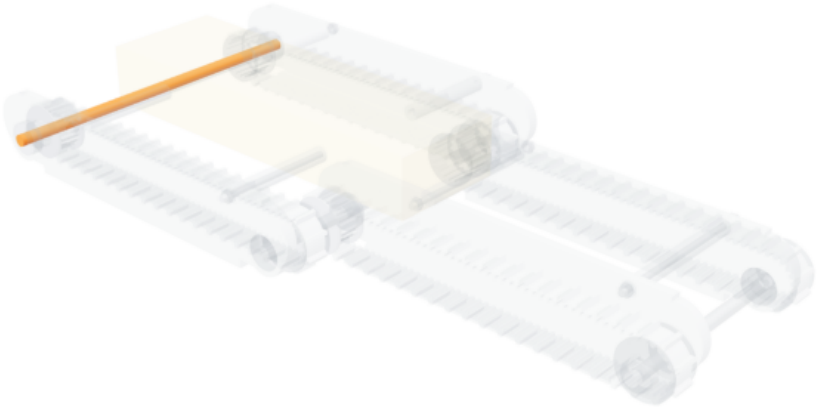
Carries only the front segment's two toothed wheels and runs in the front plates' bearings. It is not driven; it just spins with the track. Shorter than the others because the front plates are inboard.

Where to get it

Cut from the same Ø8 rod as the middle axle (one 400 mm length gives 230 + 160).

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-04 in the package PDF.

Rear axle



Quantity: 1

BUY rod, cut to 230 mm

What it does

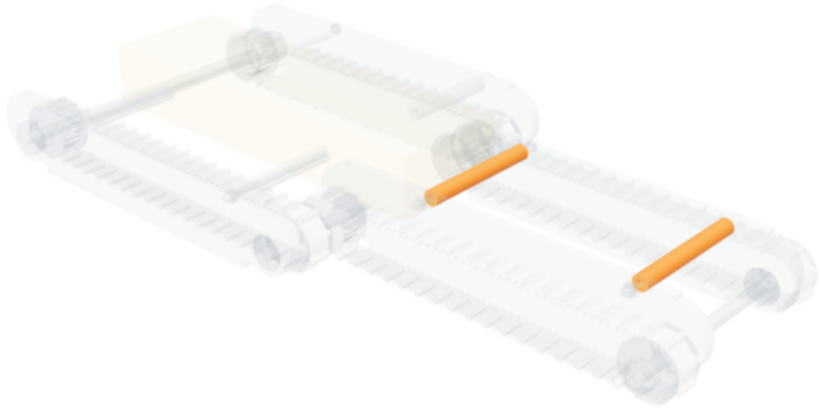
Carries the rear segment's two toothed wheels and runs in the rear plates' bearings. Not driven. Reaches the outboard rear plates, so it is as long as the middle axle.

Where to get it

Cut from the second 400 mm Ø8 rod (230 + 120 for the hinge cross shaft).

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-04 in the package PDF.

Spacer, front segment, 82 mm



Quantity: 2 (1 per cross bar)

MAKE: 3D print

What it does

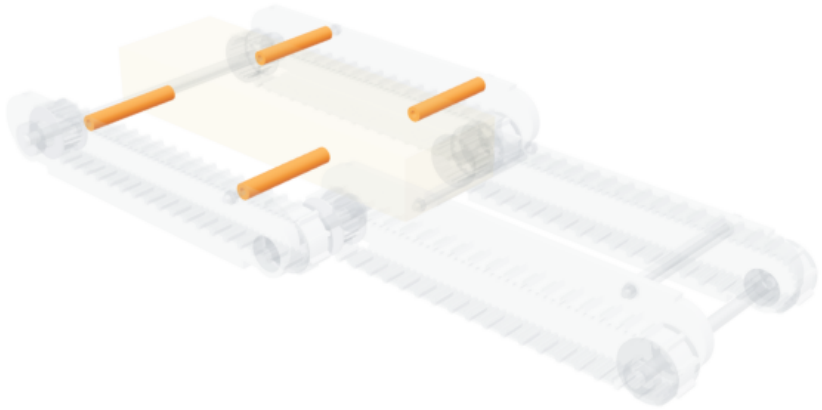
A tube that slides over the M4 rod between the two front plates. When the nuts are tightened, the plates are pulled onto the ends of this tube, so its length (82 mm) sets exactly how far apart the plates sit. Without it the nuts would squash the plates inwards onto the tracks.

Where to get it

Fab Lab, PETG, standing on end, 20 min each. File stl/spacer_front_x2.stl. Or cut from Ø10 aluminium tube with a 4.5 mm bore.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-05 in the package PDF.

Spacer, rear segment, 65 mm



Quantity: 4 (2 per cross bar)

MAKE: 3D print

What it does

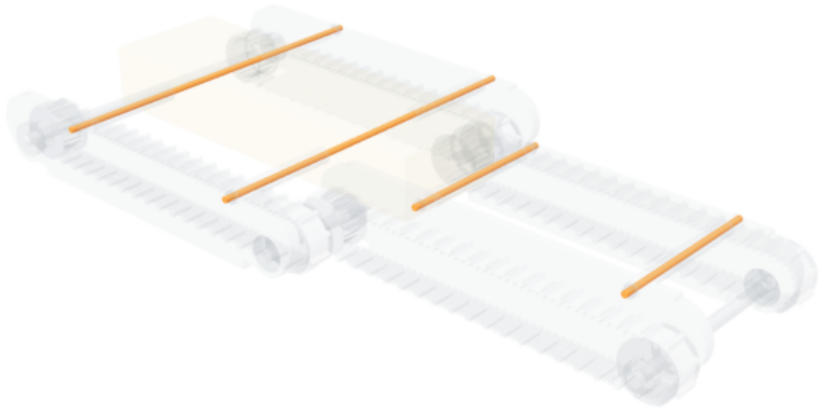
Same job for the rear segment, but the box sits in the middle of the gap, so each rod carries two shorter tubes, one each side between the box wall and the plate. Plate - spacer - box - spacer - plate is clamped as one by the nuts, and that is what holds the box on.

Where to get it

Fab Lab, PETG, file stl/spacer_rear_x4.stl, or Ø10 aluminium tube cut to 65 mm.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-05, TS-08 in the package PDF.

Cross bar: M4 threaded rod



Quantity: 4 (2 x 110 mm front, 2 x 238 mm rear)

BUY: RS Singapore or hardware shop

What it does

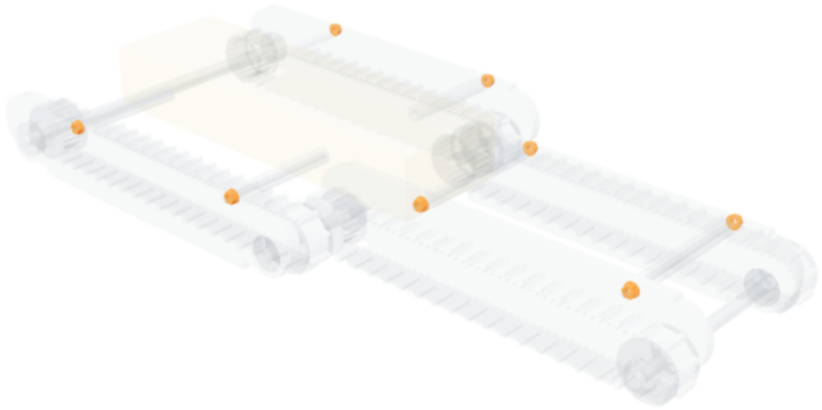
The bolt that ties the two plates of a segment together through the spacers. On the rear segment it also passes through the box walls and is the only thing attaching the rear segment to the box. Nut and washer at each end.

Where to get it

sg.rs-online.com, RS PRO stainless M4 x 1 m, stock no. 200-9958 (sold as a pack of 5, S$49.69 ex GST), or one M4 x 1 m rod from a hardware shop on the vendor list for about S$3. One rod gives all four pieces (696 mm).

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-00, TS-08 in the package PDF.

M4 nuts and washers



Quantity: 8 + 8

Design Lab spares, else Element14 SG

What it does

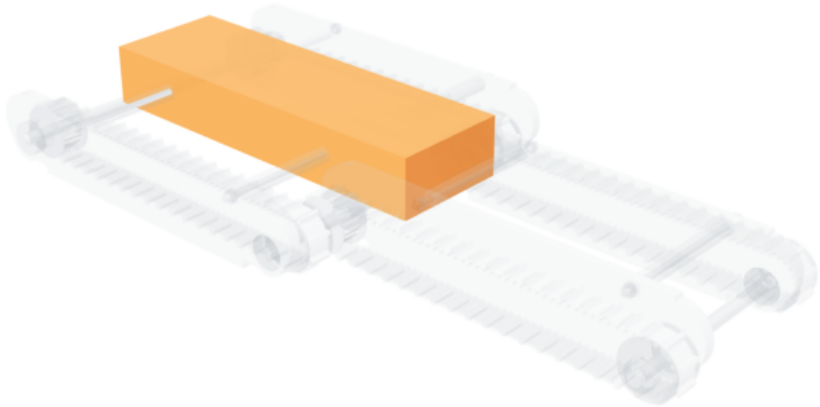
One nut and one washer on each end of each cross bar rod. Tightening them clamps plates, spacers (and the box, on the rear segment) together.

Where to get it

Design Lab (EA-03-04) spares first. Else sg.element14.com > hex nuts, M4 pack of 100 about S$6.46, and M4 washers; 1-2 business days. Also needed from the same place: 6 x M3 x 14 bolts + nuts for the hinge sector gears, and 16 x M4 set screws if MISUMI does not supply them.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-00 in the package PDF.

Box (team's part, shown for reference)



Quantity: 1

BUY an enclosure, or laser-cut at the Fab Lab

What it does

Holds the drive motor, hinge motor, driver, battery and the platform mount. It sits between the rear plates, 80 mm wide, from 215 mm behind the middle axle to 40 mm in front of it, and is clamped by the two rear cross bars. It must stop at 37 mm from the centre near the middle axle (for the sector gear) and must not reach past 50 mm forward of it (hinge swing).

Where to get it

Any ABS project enclosure about 250 x 80 x 50 mm (RS SG or Shopee, S$10-20), drilled with 2 x Ø4.5 holes per long wall at 60 and 200 mm from the rear end, 28 mm above the axle line. Or a finger-joint acrylic box from makercase.com cut on the Fab Lab laser.

Orange = this part; everything else faded. 1:5 model. Drawing sheet TS-08 in the package PDF.